



The timing of the involvement of the frontal eye fields and posterior parietal cortex in visual search

Roger Kalla^{a,*}, Neil G. Muggleton^{a,*}, Chi-Hung Juan^{c,d}, Alan Cowey^b and Vincent Walsh^a

^aInstitute of Cognitive Neuroscience and Department of Psychology, University College London, ^bDepartment of Experimental Psychology, University of Oxford, UK, ^cInstitute of Cognitive Neuroscience, National Central University, Jhongli and ^dLaboratories for Cognitive Neuroscience, National Yang-Ming University, Taiwan

Correspondence to Neil Muggleton, Institute of Cognitive Neuroscience and Department of Psychology, University College London, Alexandra House, 17 Queen Square, London WC1N 3AR UK

Tel: + 44 (0)20 7679 1160; fax: + 44 (0)20 7813 2835; e-mail: n.muggleton@ucl.ac.uk

*Roger Kalla and Neil G. Muggleton contributed equally to the study.

Received 17 March 2008; accepted 15 April 2008

The frontal eye fields (FEFs) and posterior parietal cortex (PPC) are important for target detection in conjunction visual search but the relative timings of their contribution have not been compared directly. We addressed this using temporally specific double pulse transcranial magnetic stimulation delivered at different times over FEFs and PPC during performance of a visual search task. Disruption of performance was earlier (0/40 ms) with FEF stimulation

than with PPC stimulation (120/160 ms), revealing a clear and substantial temporal dissociation of the involvement of these two areas in conjunction visual search. We discuss these timings with reference to the respective roles of FEF and PPC in the modulation of extrastriate visual areas and selection of responses. *NeuroReport* 00:000–000 © 2008 Wolters Kluwer Health | Lippincott Williams & Wilkins.

Keywords: frontal eye fields, posterior parietal cortex, timing, transcranial magnetic stimulation, visual search

Introduction

The ability to locate a specific object in a cluttered scene (visual search) requires the contribution of a network of brain areas including visual, parietal, and frontal cortices. Imaging studies show that for targets defined by a conjunction of features, 'bold' response changes occur in the posterior parietal cortex (PPC) and the frontal eye fields (FEF), as well as other areas [1]. Transcranial magnetic stimulation (TMS) studies, complementing the findings in patients with brain lesions for example see Ref. [2], have shown that stimulation delivered over both PPC and FEF disrupts performance on this type of search task [3–6] whereas search for a target defined by a single attribute (such as colour) remains unaffected. Elevation of response times and reduction of sensitivity owing to transcranial magnetic stimulation (TMS) has been seen for both stimulation sites, with the type of effect depending on task configuration. If large search arrays are used, elevated response times are typically seen [3,6,7]. When small, briefly presented stimuli are used, effects are more likely to be seen on sensitivity measures [6]. The effects of TMS over PPC or FEF on visual search performance can also differ as a function of task configuration, response mode or dependent variable (cf Ref. [8]). This means that so far, in humans at least, the relative contribution of these two brain regions to successful visual search remains to be evaluated and to do this requires that they are tested under identical stimulus and response conditions, which to date has not occurred.

Dissociation of the roles of these two areas is particularly important because of the degree of overlap of some of the processes typically ascribed to them, for example, both have been suggested to have representations of saliency maps with respect to a search target [9] and both have been associated with a top down modulation of extrastriate cortex [10].

The temporal domain may be one in which differences between the contributions of FEF and PPC to visual search might arise and this is a useful place from which to start to probe differential contributions of the two regions. O'Shea *et al.* [5] found that the disruption of conjunction visual search performance with FEF TMS occurred only when the pair of pulses were delivered 40 and 80 ms after the onset of a search array. This is consistent with the response latencies of FEF neurons within the 40–80 ms range [11]. These early response times, together with anatomical investigations of the connectivity of FEF within the visual hierarchy and the reports of FEF neurons showing selectivity for visual features and targets [9] has led to the suggestion that FEF maybe considered as part of an early or fast stage of visual processing in addition to its traditional postperceptual role in programming saccades [12,13].

In contrast, the PPC involvement in visual search performance has been reported to be at 100 ms (target present) and 160 ms (target absent) after array onset in humans [3]. Furthermore, Ellison *et al.* [8], showed that

human PPC is important for the visuomotor components of a visual search rather than the purely visual components. Additionally, location specific target processing in search tasks is associated with lateralized electrophysiological changes in the parietal lobe (the N2pc component) around 170 ms after presentation of a search task [14]. For conjunction search, this component is disrupted by TMS delivered over parietal cortex 100 ms after presentation of the array [15]. Taken together, these diverse lines of enquiry suggest that the contribution of the FEF to visual search may precede that of PPC.

Despite the availability of data from TMS studies in humans regarding the timing of involvement of both FEF and PPC in visual search, the data for the two areas is not directly comparable. When FEF timing was investigated small, briefly presented stimuli were used and the dependent measure used was sensitivity, whereas large arrays have been used in the PPC studies with the dependent measure being response times. Although these studies are suggestive of a temporal dissociation, it is also possible that such a difference reflects, for example, differences in stimulus presentation or dependent variable altering the processing time required to perform the task (either owing to differences in difficulty or duration of the presentation of visual information). Additionally, eye movements could occur freely in the PPC experiment but fixation was maintained in the FEF study and this difference alone could account for the results.

Accordingly, in this study, we directly compared the timing of the involvement of PPC and FEF in visual search using d-prime as a measure, employing small search arrays and excluding eye movements. The same participants received TMS to both of the stimulation sites on different occasions to facilitate comparison of the timing of the involvement of these areas in visual search.

Unlike the traditional hierarchical scheme of the visual system, which places FEF higher in the hierarchy than PPC, data from single cell recordings suggest much earlier responses for FEF than PPC (see Ref. [11]). We therefore predicted that, consistent with the pattern of timings seen in the studies by O'Shea *et al.* [5] and Ashbridge *et al.* [3], disruption would occur at an earlier stimulation time when TMS was delivered over FEF than over PPC.

Methods

Participants

Nine participants, (6 males, 3 females, mean age 29.2 ± 4.9 years, all right handed) took part. Seven of them had prior experience of TMS. All gave informed consent before taking part. The experiment conformed to the Declaration of Helsinki, and was approved by the University College London ethics committee.

Task

The visual search task consisted of a small search array (2° by 2°) presented centrally on a PC monitor (refresh rate 100 Hz), at a distance of 57 cm from a chin rest. The array was divided into a 6×6 invisible grid with each of 10 elements (0.2° by 0.2°) assigned to a grid 'cell' at random and then located at random within the cell. Array elements were all triangles that could point either up or down and be either orange (Commission Internationale de l'Éclairage $x=0.480$, $y=0.384$) or purple (Commission Internationale

de l'Éclairage $x=0.249$, $y=0.195$) and were matched for luminance (71 cdm^{-2}). Each participant performed the task on two occasions (with sessions carried out on different days). The target type differed on each occasion of testing (e.g. for the first session the target may have been orange and pointing upwards and on the second purple and pointing downwards). The target element was present for 50% of trials and on these occasions one less distractor of the same colour was presented such that there was always the same number of elements in target present and target absent arrays.

A trial consisted of a fixation cross (400–700 ms) followed by presentation of the search array for a duration determined for each participant (see below) which was then immediately masked until a response was made using a pattern mask with the same dimensions as the search array (Fig. 1). Participants were required to indicate the presence or absence of the target by a key press. The duration of presentation of the stimulus was first determined for each participant using a Bayesian adaptive thresholding method [16]. Briefly, this method consisted of blocks of 40 trials in which the algorithm appropriately adjusted the presentation duration after each response to produce 75% accuracy (a level of performance expected to be equivalent to a d-prime of 1) at the end of the block. Blocks were repeated until the threshold varied by less than 20 ms on successive blocks. The lowest threshold was then rounded down to the nearest 10 ms and this duration was used for the experimental blocks. The mean thresholds were 223 ms for FEF sessions and 220 ms for PPC sessions (see below).

Each experimental block consisted of 40 trials. In each block pairs of TMS pulses, with 40 ms between the pulses, were delivered over the site being tested in that session. Pulses separated by 40 ms were used because they maintain temporal resolution while having effects greater than single pulse TMS owing to summation of their effect on the cortex [5,17]. The advantage of using double pulse TMS can be seen if one considers the increase in the number of trials, if

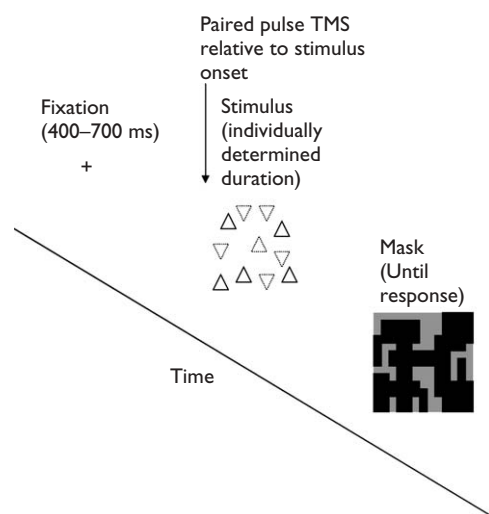


Fig. 1 Time course of a typical trial. TMS was delivered with respect to the onset of the search array. Trials were terminated if an eye movement occurred between the onset of the fixation and the appearance of the mask. TMS, transcranial magnetic stimulation.

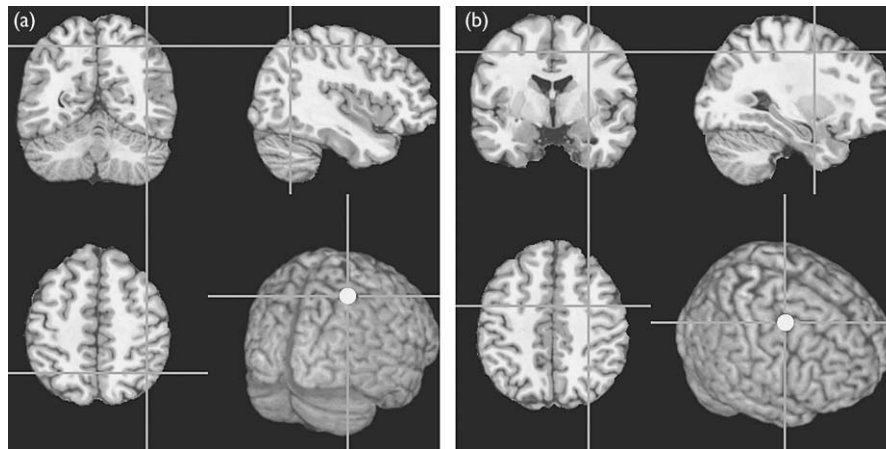


Fig. 2 Stimulation sites. (a) Posterior parietal cortex and (b) frontal eye field sites of stimulation are shown for one of the participants. Sites were marked on individual anatomical scans from transformed coordinates. The PPC site lies above the lower bank of the intraparietal sulcus. The FEF site is above the posterior middle frontal gyrus, just in front of the junction of the precentral sulcus and the superior frontal sulcus. FEF, frontal eye field; PPC, posterior parietal cortex.

one wished to sample a 200 ms poststimulus onset processing window of 200 ms as we do here. We used 6 timings of double pulse TMS. Five of these were related to the onset of the stimulus with the first/second of the pair of pulses occurring at 0/40, 40/80, 80/120, 120/160, or 160/200 ms after onset of the search array. The final timing was with the first pulse of the pair occurring concurrent with the onset of the mask. Timings were blocked such that all 40 trials of a block were presented with the same pulse timings. Two blocks were presented for each timing and two further blocks with no TMS were also presented. The order of blocks was randomized for each participant. Half of the participants received TMS over PPC during the first testing session and TMS over FEFs for the second session and half the reverse order. Additionally, the target in the search array was also balanced across participants such that the target types were evenly distributed with respect to the order in which the sites were stimulated (Fig. 2).

Eye monitoring

To ensure the data obtained were not confounded by either blinks or saccades, eye position was monitored during the experiment. This was achieved by presenting the task using a SMI Eyelink. The task was programmed in C++ (MS Visual Studio 6) and utilized the Eyelink software development kit. The eyes were monitored from the onset of the fixation cross until the presentation of the mask. If, during this time, any blinks or eye movements were detected the trial was terminated and any response data discarded. The rejection rate was 4.1% for FEF stimulation and 5.0% for PPC stimulation.

TMS and site localization

A Magstim 200 Super-Rapid Stimulator was used to deliver TMS at 60% of maximum machine output (approximately 1.2 T, duration 1 ms) over PPC and FEF. A fixed stimulation level was used because it has proven successful and replicable in many studies and over a wide range of tasks [3,6,18] and because motor cortex excitability does not provide a good guide to TMS thresholds in other cortical areas [19]. Stimulation was delivered via a 70 mm figure of

eight coil held clamped in position with the handle at approximately 45° to the midline and pointing in a medial to lateral/anterior to posterior direction (with the direction of the current over the stimulation site travelling in the same medial to lateral/anterior direction).

Sites were localized in each participant using a frameless stereotaxy system (Brainsight, Rogue Research, Montreal, Canada). Each participant was coregistered with their own structural MRI scan, on which the FEF and PPC locations had been identified, and the scalp positions overlying the sites to be stimulated were marked on a cloth swimming cap which was worn throughout the test session. Briefly, identification of sites on the structural scans was achieved by the following procedure. Individual MRIs were normalized against a standard template using the FSL software package (FMRIB, Oxford Centre for Functional MRI of the Brain, University of Oxford, Oxford, UK). This resulted in a matrix, which described the transformation applied to the structural scan to produce the normalized brain. This was then reverse-applied to the coordinates for FEF (31, -2, 47, [20]) and PPC (42, -58, 52, [18,21]) to obtain the location of these sites in the original structural scan. These locations were then marked on the MRI scans in the Brainsight system.

Results

Accuracy data from each stimulation condition for each site were collected and used to calculate d-prime scores (and the bias value, c) (see Fig. 3 for d-prime scores obtained). As either zero misses or zero false alarms render d-prime incalculable, in cases where this occurred a correction of half a response of the appropriate kind was employed.

D-prime scores were analysed using repeated measures analysis of variance with factors of TMS site (FEF, PPC) and TMS time (0/40, 40/80, 80/120, 120/160, 160/200 ms, mask onset, no TMS). No significant main effect of TMS [$F(1,8)=0.006$, $P=0.0939$] or of time [$F(1,8)=0.201$, $P=0.201$] exists but there was a highly significant interaction [$F(6,48)=7.328$, $P=0.004$]. This interaction was investigated by comparison of each TMS time with the no TMS condition

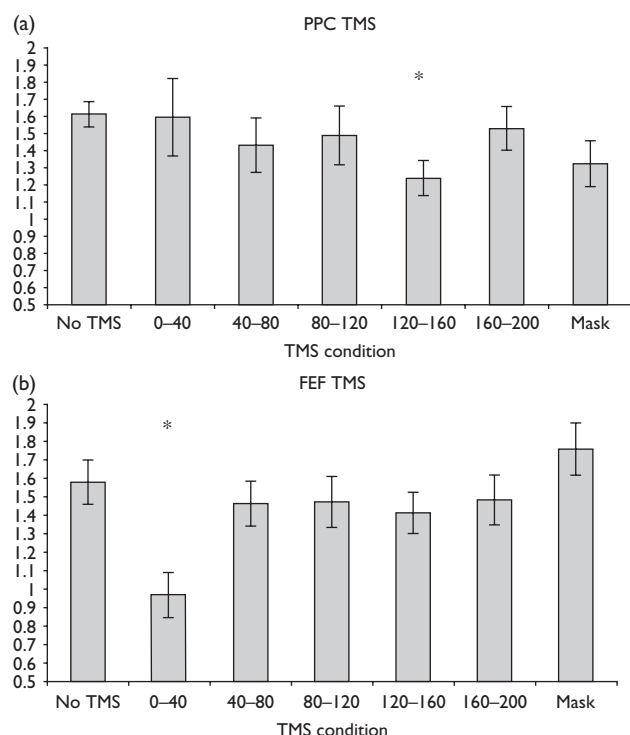


Fig. 3 Results. D-prime scores were calculated for each time of delivery of TMS for TMS over (a) posterior parietal cortex and (b) frontal eye fields. * indicates significant effects of TMS. FEF, frontal eye field; PPC, posterior parietal cortex; TMS, transcranial magnetic stimulation.

for each site (paired *t*-tests, corrected for multiple comparisons). This revealed that performance was disrupted by the FEFs TMS pulse pair delivered 0/40 ms after array ($t=3.908$, $P=0.014$) onset whereas PPC TMS disrupted performance only with the pulse pair delivered 120/160 ms after array onset ($t=4.909$, $P=0.003$). These were the only two comparisons to reach significance.

Repeated measures analysis of variance with factors of TMS site and TMS time revealed no significant effects on either bias (i.e. proportions of absent/present judgements) or response times for either present or absent trials (response times were generally slightly higher for absent judgements but this difference was also not significant).

Discussion

We compared the timing of the involvement of the PPC and FEFs in visual search for a target defined by a conjunction of two features, a task in which both areas have previously been shown to be involved using a range of methodologies [1,3,6,7,9]. The comparison was facilitated by matching the initial accuracy on the task for all participants (by means of an adaptive thresholding procedure), and the site of stimulation being a within-subject factor. The pattern of data obtained was consistent with the initial hypothesis that the time at which FEF stimulation would disrupt performance would be earlier than that for PPC stimulation, with TMS over FEF disrupting performance when delivered at 0 and 40 ms and TMS over PPC having a significant effect much later, at 120 and 160 ms.

The use of the same task design with stimulation over FEF and PPC employed here shows a clear difference in the timing of the involvement of these two areas in conjunction search performance. FEF TMS resulted in disruption of performance when pulses were delivered at 0 and 40 ms after onset of the search array. This window overlaps with that previously reported by O'Shea *et al.* [5] who observed effects with pulses delivered at 40 and 80 ms poststimulus onset. Although the effect seen here was earlier than previous studies, it is worth noting that the 40/80 ms, despite not showing a significant reduction in *d*-prime, did show a reduction for 7 of the 10 participants, more than any apart from the 0/40 ms stimulation. Consequently, the different time point seen here may be a result of interparticipant variability. Taken with the findings of previous studies, the data suggest that the effective timing for FEF interference lies within the first 80 ms after stimulus onset.

TMS delivered over PPC disrupted performance when the pulses occurred at 120 and 160 ms after stimulus onset, confirming that this area is involved in task performance at a later time than FEF. This window of effect is consistent with previous findings [3], despite the differences in the dependant measure and the task configuration. It would seem beneficial in future studies, which investigate the roles of these two areas in tasks such as visual search to evaluate the extent of their contribution to visual and motor/response selection elements of performance and the dissociability of these [8]. For example, dissociation of visual and motor processes in the FEFs has been reported in microstimulation studies in macaques [12] with analogous findings using TMS and a similar task [22]. Although saccades were not required in this task (and trials where they occurred were excluded), the results support the argument that the FEF effect seen here is a consequence of disruption of a visual process, whereas the TMS effect over PPC is more likely to be a consequence of an effect on response selection or visuo-motor transformation [8]. The early involvement of FEF may reflect either a role for this area in modulating the sensitivity of visual areas such as V4 and other extrastriate regions to target relevant information [10,23] or a role in spatial priming of an area to which a saccade may be prepared. The later role of the PPC may also be consistent with modulatory effects of visual cortex activity but the later timing together with the finding that PPC, unlike FEF, is equally important for target absent trials, suggests to us that the effect is more likely to be due to interfering with visuomotor transformation required for the manual response [8].

The comparison of the PPC and FEFs has many further aspects that we are currently investigating and that are important to a description of these areas' function in visual functions. First, it is important to know how these areas differ in their contribution according to mode of response. Here we used a key press and participants maintained eye fixation but our preliminary and other [8] results clearly suggest that the contribution of areas may be affected by response mode. It is important that we compare the roles of these areas during free eye movements and saccade localization. Second, as shown by Bjoertomt *et al.* [21] the role of an area in a visuospatial task depends on whether that task occurs in near or far space – a comparison that remains to be tested with FEF TMS. Third, the spatial extent of the array is an important variable as shown by Ellison *et al.*

al. [8] whose data challenge the notion that PPC is important for conjunction analysis *per se* by showing that TMS over PPC only interferes with conjunction target identification if the location of the stimulus to be identified is unknown. Fourth, as shown by Ellison *et al.* [8], PPC seems only to be involved in unfamiliar conjunction searches, a factor not yet tested with FEF TMS. Finally, several studies of human FEF [23–25] have now shown that they are important in top down control functions usually attributed to PPC. At our current state of knowledge, then, there is a great deal of overlap in our descriptions of FEF and PPC functions and to begin to identify the extent to which their functions overlap and differ requires a careful, step by step comparison of the two areas under identical conditions.

Conclusion

The contribution of the PPC and the FEFs to performance of identical conjunction visual search tasks can be dissociated temporally. FEF disruption occurs at an earlier time point than PPC disruption, consistent with the timing of the responses of these areas seen in macaque studies [15]. Further differences between the processes carried out by these two areas during visual search in humans remain to be elucidated.

Acknowledgements

Support: Neil G. Muggleton, Vincent Walsh and Alan Cowey were funded by a Wellcome Trust Project grant. V. Walsh was also funded by the Royal Society and R. Kalla was funded by the DFG. C-H. Juan was funded by the National Science Council, Taiwan (NSC-96-2413-H-008-001-MY3). This study was also supported by a Wellcome Trust Equipment award to the Institute of Cognitive Neuroscience.

References

- Donner TH, Kettermann A, Diesch E, Ostendorf F, Villringer A, Brandt SA. Visual feature and conjunction searches of equal difficulty engage only partially overlapping frontoparietal networks. *Neuroimage* 2002; **13**:16–25.
- Eglin M, Robertson LC, Knight RT. Cortical substrates supporting visual search in humans. *Cerebral Cortex* 1991; **1**:262–272.
- Ashbridge E, Walsh V, Cowey A. Temporal aspects of visual search studied by transcranial magnetic stimulation. *Neuropsychologia* 1997; **35**:1121–1131.
- Ellison A, Cowey A. Time course of the involvement of the ventral and dorsal visual processing streams in a visuospatial task. *Neuropsychologia* 2007; **45**:3335–3339.
- O'Shea J, Muggleton NG, Cowey A, Walsh V. Timing of target discrimination in human frontal eye fields. *J Cogn Neurosci* 2004; **16**:1060–1067.
- Muggleton NG, Juan C-H, Cowey A, Walsh V. Human frontal eye fields and visual search. *J Neurophysiol* 2003; **89**:3340–3343.
- Walsh V, Ellison A, Ashbridge E, Cowey A. The role of the parietal cortex in visual attention – hemispheric asymmetries and the effects of learning: a magnetic stimulation study. *Neuropsychologia* 1999; **37**:245–251.
- Ellison A, Rushworth M, Walsh V. The parietal cortex in visual search: a visuomotor hypothesis. *Suppl Clin Neurophysiol* 2003; **56**:321–330.
- Bichot NP, Thompson KG, Chenthal Rao S, Schall JD. Reliability of macaque frontal eye field neurons signalling saccade targets during visual search. *J Neurosci* 2001; **21**:713–725.
- Moore T, Fallah M. Control of eye movements and spatial attention. *Proc Natl Acad Sci U S A* 2001; **98**:1273–1276.
- Bullier J. Integrated model of visual processing. *Brain Res Rev* 2001; **36**:96–107.
- Sato TR, Schall JD. Effects of stimulus-response compatibility on neural selection in frontal eye field. *Neuron* 2003; **38**:637–648.
- Juan CH, Shorter-Jacobi SM, Schall JD. Dissociation of spatial attention and saccade preparation. *Proc Natl Acad Sci U S A* 2004; **101**:15541–15544.
- Hopf JM, Boelmans K, Schoenfeld MA, Luck S, Heinze HJ. Attention to features precedes attention to locations in visual search: evidence from electromagnetic brain responses in humans. *J Neurosci* 2004; **24**:1822–1832.
- Fuggetta G, Pavone E, Walsh V, Kiss M, Eimer M. Cortico-cortical interactions in spatial attention: a combined ERP/TMS study. *J Neurophysiol* 2004; **95**:3277.
- Kontsevich LL, Tyler CW. Bayesian adaptive estimation of psychometric slope and threshold. *Vision Res* 1999; **39**:2729–2737.
- Juan C-H, Walsh V. Feedback to V1: a reverse hierarchy in vision. *Exp Brain Res* 2003; **150**:259–263.
- Goebel S, Walsh V, Rushworth MF. The mental number line and the human angular gyrus. *Neuroimage* 2001; **14**:1278–1289.
- Stewart LM, Walsh V, Rothwell JC. Motor and phosphene thresholds: a transcranial magnetic stimulation correlation study. *Neuropsychologia* 2001; **39**:415–419.
- Paus T. Location and function of the human frontal eye-field: a selective review. *Neuropsychologia* 1996; **34**:475–483.
- Bjoertomt O, Cowey A, Walsh V. Spatial neglect in near and far space investigated by repetitive transcranial magnetic stimulation. *Brain* 2002; **125**:2012–2022.
- Juan C-H, Muggleton NG, Tzeng OJL, Hung D, Cowey A, Walsh V. Segregation of visual selection and saccades in human frontal eye fields. *Cereb Cortex* 2008; doi:10.1093/cercor/bhn001.
- Silvanto J, Lavie N, Walsh V. Stimulation of the human frontal eye fields modulates sensitivity of the human visual cortex. *J Neurophysiol* 2006; **96**:941–945.
- Taylor P, Nobre A, Rushworth MFS. FEF TMS affects visual cortical activity. *Cereb Cortex* 2007; **17**:391–399.
- Grosbras MH, Paus T. Transcranial magnetic stimulation of the human frontal eye field facilitates visual awareness. *Eur J Neurosci* 2003; **18**:3121–3126.

AUTHOR QUERY FORM

**LIPPINCOTT
WILLIAMS and WILKINS**

JOURNAL NAME: WNR

ARTICLE NO: 4479

QUERIES AND / OR REMARKS

[illegible]